Spatial–Spectral Similarity Based on Adaptive Region for Landslide Inventory Mapping With Remote-Sensed Images

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***Abstract*— Landslide is one of the most serious geological disasters around the world, and acquiring landslide inventory mapping (LIM) with remote-sensed images (RSIs) plays an important role in disaster relief. However, various external imaging conditions of bitemporal RSIs usually cause pseudo-changes and challenges for achieving satisfied LIMs. In this article, a pioneering change magnitude measured distance named Spectral–Spatial Similarity based on Adaptive Region (S^3AR) is proposed for achieving LIMs with bitemporal RSIs. First, an adaptive region is proposed to utilize the spatial-contextual information around each pixel, because the shapes and size of a landslide site are usually irregular and unpredictable. Then, a shape description algorithm is proposed for constructing a shape description vector, which aims at measuring the spatial difference of adaptive regions. Finally, to improve the separability between the landslide area and the background, brightness is suggested to couple with the shape description vector of an adaptive region to generate spatial–spectral similarity to measure the change magnitude between pairwise adaptive regions from the bitemporal RSIs. When the entire bitemporal RSIs are scanned and calculated via these steps, a change magnitude image between bitemporal RSIs can be generated, and then binary LIMs are obtained by a binary threshold. Experiments based on comparing eight state-of-the-art approaches demonstrated the feasibility and superiorities of the proposed S^3AR for achieving LIMs with bitemporal RSIs. For example, the improvements on the four datasets are 5.81%, 14.06%, 6.03%, and 20.51% in terms of total error.**

***Index Terms*— Adaptive region, landslide inventory mapping (LIM), remote sensing images, spectral-spatial similarity.**

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I. Introduction

LANDSLIDE is a common natural disaster around the world. Large amounts of economic loss and human casualties are caused by landslides in each country [1]. For example, in the USA, some research surveys indicate that landslides cause an average of 25–50 fatalities annually and contribute to economic losses of billions of dollars each year [2]. In Canada, landslides cause losses exceeding 281 – 450 million per year, and from 2007 to 2015, a total of 5228 fatalities were caused by landslide events in the Indian Himalayan region [3], [4]. Thanks to the advantages of remote sensing technology, landslide inventory mapping (LIM) can be achieved with remote-sensed images (RSIs), SAR images and so on [5], [6], [7]. LIM playing an important role in landslide site management and landslide disaster relief [8], [9], [10].

Many studies have indicated that change detection with bitemporal RSIs can achieve landslide inventory maps (LIMs) for practical applications [11], [12], [13], [14], [15]. Here, these approaches are briefly divided into traditional methods (non-deep learning) and deep learning methods [16], [17]. Their details are reviewed below.

In traditional methods, many studies have indicated that exploring spatial-contextual features around a pixel can improve change detection performance with RSIs [18], [19], [20], [21]. One of the classical spatial-contextual feature exploration methods for LIMs with RSIs is the regular window-based technique [22], [23]. For example, local spectrum-trend similarity [24], change vector analysis (CVA) fuzzy clustering mean (CVA_FCM) [25], sliding window-based extension [26], and multiscale object-based image analysis landslide detection technique [27] well demonstrated that sliding windows or multiscale windows can smooth noise and improve the visual performance and accuracies of the change detection map with RSIs. However, various ground targets usually have different sizes and shapes, such as landslides and rivers; a regular shape and size "window" may be insufficient for exploring spatial-contextual information for change detection.

In addition to window-based techniques, mathematical models are usually adopted to describe the pixel's distribution to utilize spatial-contextual information and improve change detection performance. For instance, the multiresolution level set (MLS) [28], edge-based level set evolution [29], expectation–maximization-based level set [30] are popular

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in the community of change detection with RSIs. Markov random field-based change detection (CDMRF) [31], active learning and Markov random field [32], Fuzzy C-means, and Markov random field [33] are widely used in LCCD with RSIs. In addition, Gabor-wavelet-based difference feature (GWDM) [34], Gabor-feature-based unsupervised change detection [35], Gabor-wavelet and robust semi-supervised fuzzy C-means (Semi_FCM) [36] improve change detection performance via smoothening the noise of RSIs in a multiscale way. Many studies for LCCD with RSIs well demonstrated that mathematical models are feasible and effective for improving change detection performance and accuracies with RSIs. However, the approaches based on mathematical models usually required parameters and strict mathematical models are insufficient to cover various ground targets with different shapes and sizes.

CVA is also a very popular technique in the field of LCCD with RSIs [37], [38]. These methods include multiscale morphological compressed CVA [39], CVA_FCM [25], and robust CVA [40], which focused on using different channel information to separate the detection ground targets from the background. To further improve the performance of LIM and to cover the various ground targets, the idea of adaptation was applied in many methods. For example, modified seed-region growing [41], object-oriented key point vector distance [42], and multiscale segmentation [43] aim at exploring the spatial–spectral contextual information around each pixel via multiscale segmentation and superpixel. Adaptive weight is also usually coupled with some change detection approaches for utilizing spatial-contextual information in an adaptive way. Examples include self-adaptive weight CVA [44], CVA with adaptive weight Gaussian mixture model [45], and CVA with adaptive weight Gaussian mixture model [46] have been applied successfully in improving change detection performance. Similar to the method described above, Lv et al. proposed some adaptive region-based approaches for change detection with RSIs [47], and this idea has been applied to improve the performance of LIM [48], [49]. The advantage of utilizing adaptive region to explore spatial-contextual information lies in that it can fit and preserve the shape and size of various ground targets, and the details of the adaptive region-based approach can also be found in the literature [50]. In summary, a large number of studies have indicated the feasibility and effectiveness of utilizing spatial-contextual information in improving the detection performance of LIM with RSIs.

In addition to traditional methods for LIM with RSIs, deep learning approaches are very popular in the community of LIM. Deep learning neural networks have some advantages in identifying and classifying ground targets in RSIs because they can extract deep and abstract image features that play an important role in pattern recognition [12], [51], [52], [53]. In the past decade, some supervised deep-learning methods have shown excellent performance in LIM [54], [55], [56], [57]. The character of the supervised deep learning method is that the training process needs many labeled samples. For example, Fang et al. proposed a neural network that requires deep supervision for change detection (DSMANET) [58]. Su et al. [59] invented a supervised method for LIM using dual-temporal images and rasterized digital terrain models to train the neural network. Abeywickrama et al. [60] proposed a supervised method for change detection using a road network based on k-nearest neighbors (KNN). In addition to the supervised deep learning approach described above, the unsupervised deep learning methods use unlabeled data to train the model, relying on the model to discover features and patterns within the data on its own [61]. Lei et al. [62] proposed an unsupervised novel approach based on a fully convolutional network within pyramid pooling for LIM with RSIs. Ramos-Bernal et al. [63] developed an unsupervised network for LIM based on linear regression (LR), chi-square transformation, and CVA. Wu et al. [64] proposed a novel unsupervised model called kernel principal component analysis (KPCAMNET) convolution for extracting features from RSIs. The adaptive idea is also employed to construct neural networks, such as conditional attention lightweight network [65], feature-fusion segmentation network [66], and attribution deep learning interpretation model [67], where the attention mechanism calculated the soft weight to enhance the target's information or area. Although deep learning-based techniques have been applied successfully in applications with RSIs, satisfactory change detection performance of supervised deep learning networks usually depends on many training samples, the labeling of which is time-consuming and labor-intensive. An unsupervised deep learning network does not need a training sample, but the loss function designation of which is usually challenging.

In this article, due to the landslides exhibiting unpredictable sizes and irregular shapes, a new method based on spectral-shape similarity for LIM is proposed. The major contributions of this article can be summarized as follows.

1) A novel shape description vector is first proposed for describing the shape of an adaptive region around a pixel. The contribution of the proposed shape description algorithm lies in that it provides a new way to measure the shape of an adaptive region quantitatively. Shape quantitative description of an area in RSIs can be used to supplement the insufficiency of spectral reflectance when distinguished ground targets are spectrally similar but differently shaped.
2) A novel distance named spatial–spectral similarity based on adaptive region (S^3AR) is proposed to measure the change magnitude between bitemporal RSIs for LIMs. In the proposed S^3AR, the spatial-contextual information is not only utilized via an adaptive region around each pixel but also the shape description vector was suggested to couple with the brightness (spectral information) for improving the performance of acquiring LIMs.
3) Advanced landslide inventory maps can be obtained by our proposed approach with RSIs. Based on four pairs of real RSIs that depict the land cover change before and after some landslide events, several cognate change detection methods, including MLS [28], CVA_FCM [25], Semi_FCM [36], CDMRF [31], KNN [60], GWDM [34], KPCAMNET [64], and DSMANET [58], are compared with our proposed approach. The comparative results indicated that the

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proposed approach can achieve better visual performance and higher accuracies in terms of some widely used evaluation indicators.

The remainder of this paper is organized as below. Section II provides a detailed introduction to the method proposed in this article. Section III presents the experimental results, demonstrating the feasibility and advantages of the proposed method. Section IV summarizes the core ideas of this article.

II. Methodology

In this section, the proposed S^{3}AR-based approach for LIM with RSIs is presented in detail. As shown in Fig. 1, the proposed S^{3}AR-based approach is composed of three parts: 1) extending an adaptive region around each pixel; 2) defining the shape of the adaptive region; and 3) generating the change magnitude image with spatial–spectral similarity. The details are given below.

A. Extension of Adaptive Region Around a Pixel

Many studies have proved that considering the contextual information around a pixel adaptively can effectively smooth noise and obtain satisfied LIMs [48], [49], [50], [68]. The shape and size of a landslide site in an RSI are various and unpredictable. Therefore, in line with many studies, an adaptive region extension was adopted in our proposed approach for LIMs with remote sensing images. Details of the employed adaptive region extension algorithm are briefly reviewed below.

First, a pixel (P_{ij}) is anchored at the seed pixel, and the neighboring pixels around P_{ij} are denoted as P_{nei}; Second, the spectral difference between a pixel and its neighboring pixels is measured by $\|P_{ij} - P_{\text{nei}}\| \leq t_1$, where t_1 is a predefined parameter. If $\|P_{ij} - P_{\text{nei}}\| \leq t_1$ this is true, then the pixel P_{nei} can be marked as the pixel of the extended region. Third, at the same time, the total quantity of the pixels (q) that have been labeled as the extended region is counted. If $q \leq t_2$ is true, the second step will be conducted iteratively, where t_2 is a predefined parameter. This process is extended iteratively around the pixel P_{ij}, and the iteration is terminated until one of the two conditions ($\|P_{ij} - P_{\text{nei}}\| \leq t_1$ and $q \leq t_2$) is false.

The above statement shows that the extended region around a pixel has the following characteristics: 1) an extended region is composed of the pixels that are homogeneous with the central pixel in terms of spectral value and 2) the spectral homogeneous pixels are continuous in a spatial domain. Due to the extension of an adaptive region around a pixel growing gradually by comparing with the central pixels and its neighboring pixel, an adaptive region around a central pixel has an advantage in preserving the shape and sizes of a part of a ground target. Therefore, when an adaptive region around each pixel was adopted to utilize the spatial-contextual information for detecting LIMs, the shape and size of a landslide site may be well preserved.

B. Shape Description of an Adaptive Region

On the basis of the generation of an adaptive region around each pixel, how to describe quantitatively the shape of an adaptive region remains a problem. The extension of an adaptive region indicates that an adaptive region is a group of pixels that are spectrally homogeneous and spatially continuous. Intuitively, an adaptive region usually has high redundancy in terms of information expression. Therefore, in this section, we proposed an algorithm for describing the shape of an adaptive region. As presented in Fig. 2, the proposed algorithm aims to describe the shape of an adaptive region via direction lines, as shown in Algorithm 1.

Algorithm 1 Shape Description Algorithm for an Adaptive Region

Input: A pixel P_{ij} in the bi-temporal images and an adaptive region $R_{i,j}$ generated around P_{ij}, where $P_{ij} \subseteq R_{i,j}$.

Output: V_{ij}: where V_{ij} is the shape description vector of the adaptive region $R_{i,j}$ around the pixel P_{ij}.

Step-1: Define the total number of direction lines N, and the offset of each direction line is $\frac{2\pi}{N}$, $N \subseteq \{2, 4, 6, \ldots\}$;

Step-2: Detect the edge of the adaptive region from the pixel P_{ij} along the direction line $n \times \frac{2\pi}{N}$, where n denotes the number of direction lines for the $R_{i,j}$, and $n = \{1, 2, 3, \ldots\}$. The pixel on one direction line is assumed as P_x, if $\|P_{ij} - P_x\| \leq t_1$ (t_1 is a predefined parameter used for the generation of adaptive region) is true, then the probing progress along the direction is continually. The probing progress is terminated until $\|P_{ij} - P_x\| \leq t_1$ is false.

Step-3: Calculate the Euclidean distance between the pixel P_{ij} and the pixel P_x, then the distance is symbolized with v_i.

Step-4: Repeat Step-2 and step-3 along each direction line around the pixel P_{ij}, and the corresponding distance between the pixel and the end of each direction line can be denoted as $v_1, v_2, v_3, \ldots, v_n$.

Step-5: Finally, the distances $(v_1, v_2, v_3, \ldots, v_n)$ were constructed in sequence to construct the shape description vector (V_{ij}) for describing the shape of the adaptive region, $V_{ij} = \{v_1, v_2, v_3, \ldots, v_n\}$.

End

From the algorithm above, two aspects are worth noting.

1) The shape of an adaptive region can be described by the prosed shape description vector which is composed of a group of the distance between the central pixel and the pixel on the boundary of the adaptive region, it can be a sparse representation comparing with describing the shape of the adaptive region with the total pixels.
2) A larger value of the parameter N means describing the shape in more detail; by contrast, a smaller value of the parameter N means describing the shape in a coarser scale. In our proposed approach, the proposed shape description vector for each adaptive region is used to supplement the insufficiency of change detection based only on spectral information.

C. Definition of Spatial–Spectral Similarity

On the basis of the definition of the shape description vector of an adaptive region, a spatial–spectral similarity between two pairwise adaptive regions is proposed to measure the change

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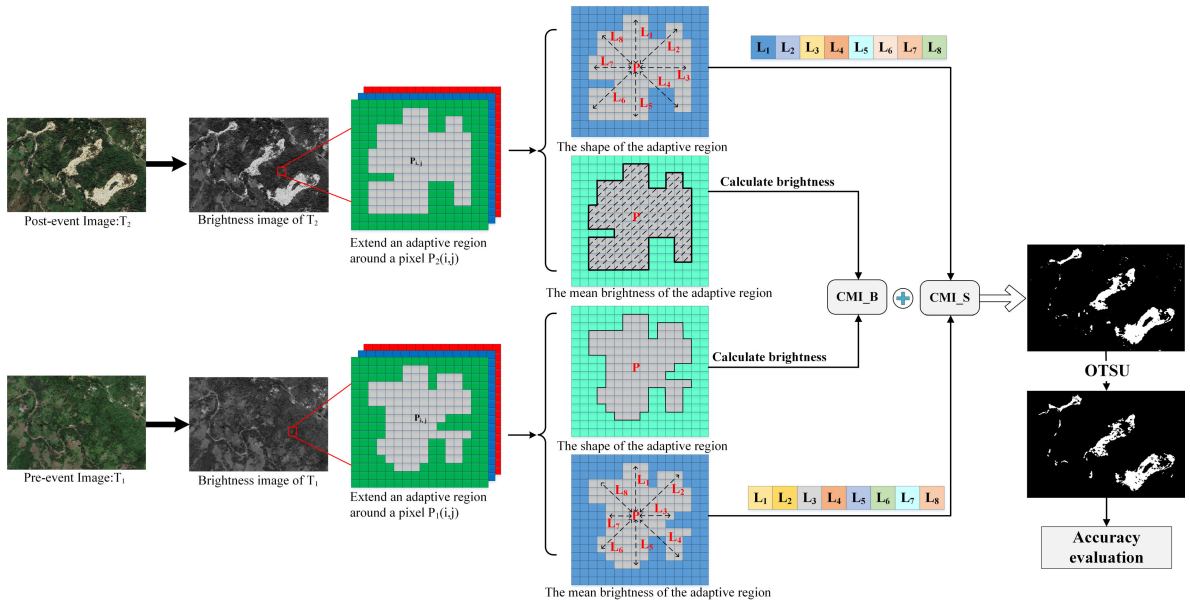


Fig. 1. Flowchart of the proposed spatial–spectral channels attention neural network.

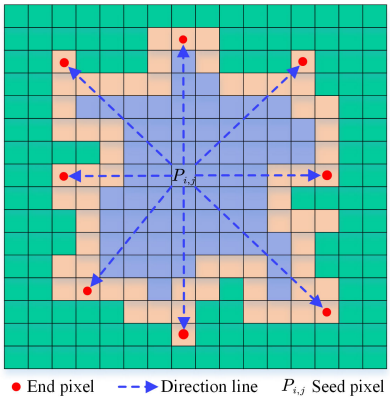


Fig. 2. Illustration of the shape description algorithm for an adaptive region ($N = 8$).

magnitude between pairwise pixels. The definition of the proposed spatial–spectral similarity is given in the following equation:

$$D_{ij} = \left\| V_{ij}^1 - V_{ij}^2 \right\| + \left\| B_{ij}^1 - B_{ij}^2 \right\|$$

where D_{ij} denotes the change magnitude between the pairwise pixels in the bitemporal images. V_{ij}^1 and V_{ij}^2 are the shape description vector for the adaptive region around the pixels at P_{ij}^1 and P_{ij}^2 in the bitemporal images ("1" and "2" denoted the bitemporal images). Here, B_{ij}^1 and B_{ij}^2 are the brightness for the pairwise adaptive regions R_1 and R_2 around the pixels at P_{ij}^1 and P_{ij}^2 in the bitemporal images, and the brightness is the average spectral value of the pixels within an adaptive region. When an entire bitemporal RSIs is scanned and measured by the proposed spatial–spectral similarity, change magnitude images between the bitemporal images can be obtained for LIMs.

In addition to existing change detection methods for achieving LIMs, the proposed approach also requires a binary threshold to partition the CMI into changed and unchanged areas. Here, an unsupervised threshold method named Otsu was used to obtain the LIMs [69]. Otsu determines a threshold for separating CMI into two classes (changed and unchanged) based on maximizing the change area's variance and minimizing the unchanged area's variance.

TABLE I
DETAIL DESCRIPTION FOR EACH DATASET

Dataset	Year	Location	Sensors	Resolution
1	2006 and 2011	Hong Kong, China	Aerial camera	0.5 m/pixel
2	2007 and 2014	Hong Kong, China	Aerial camera	0.5 m/pixel
3	2007 and 2014	Hong Kong, China	Aerial camera	0.5 m/pixel
4	2021 and 2021	Kupang, Indonesia	Google Earth	0.59 m/pixel

III. EXPERIMENT

In this section, two experiments with four pairwise remote sensing images were conducted to verify the feasibility and performance of the proposed S^{3}AR method for achieving LIM. The first experiment employed eight state-of-the-art techniques to demonstrate the advantages of the proposed method. Then, the second experiment observed the relationship between the parameters of the proposed approach and the detection accuracies. The experimental details are presented below.

A. Description of Datasets

Four pairwise datasets were adopted to validate the performance of the proposed method in the experiments, and the details of the used dataset are summarized in Table I. As shown in Fig. 3, each dataset is composed of pre-event images, post-event images, and the ground reference map. Datasets #1, #2, and #3 describe landslide events occurring in

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TABLE II
PARAMETER SETTING FOR EACH METHOD AND DATASET

Method	Dataset A	Dataset B	Dataset C	Dataset D
MLS [28]	L=2 μ=0.1	L=2 μ=0.2	L=2 μ=0.1	L=3 μ=0.3
CVA_FCM [25]	-	-	-	-
Semi_FCM [36]	C=2,T=200 $\varepsilon=10^{-5}$	C=2,T=180 $\varepsilon=10^{-5}$	C=3,T=220 $\varepsilon=10^{-5}$	C=2,T=200 $\varepsilon=10^{-5}$
CDMRF [31]	λ=50 M=5	λ=50 M=5	λ=50 M=5	λ=50 M=5
KNN [60]	-	-	-	-
GWDM [34]	W=2	W=2	W=2	W=6
KPCAMNet [64]	SN=150	SN=150	SN=150	SN=150
DSMANet [58]	-	-	-	-
S^{3}AR_OTSU	T_1=90 T_2=150 N=8	T_1=80 T_2=150 N=4	T_1=150 T_2=200 N=8	T_1=50 T_2=150 N=8

Lantau Island in Hong Kong, China. They were captured by an airplane equipped with an aerial camera, producing aerial orthoimages with a spatial resolution of 0.5 m/pixel. Dataset #4 depicts the landslide events occurring in the Kupang area in Indonesia, and it was derived from a publicly accessible landslide dataset named GVLM [70]. The GVLM dataset aims at detecting landslides with the Google Earth image service, and the bitemporal image for depicting landslides has a spatial resolution of 0.59 m/pixel.

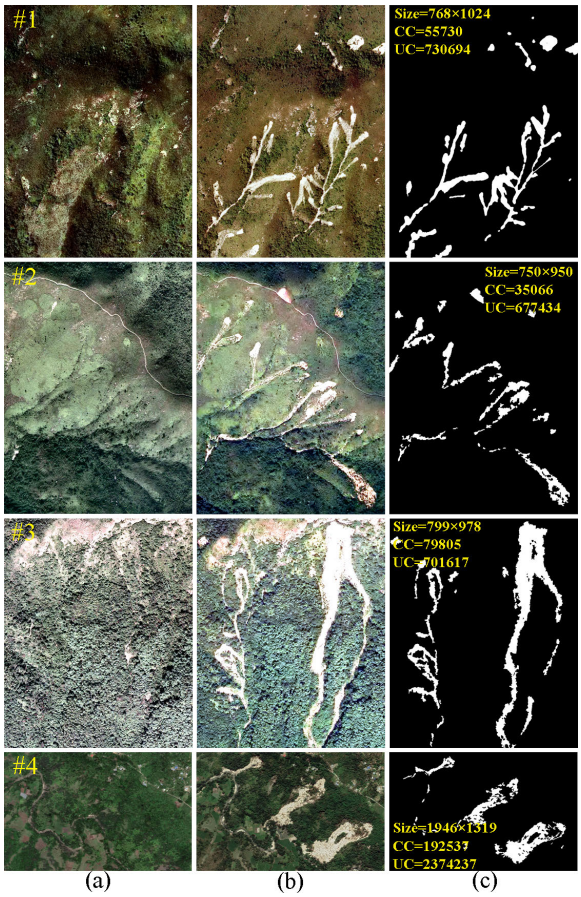


Fig. 3. Dataset for each area and their reference maps. (a) Pre-event image. (b) Post-event image. (c) Ground reference map.

B. Experimental Setup

In this section, to verify the effectiveness of the proposed method, eight state-of-the-art approaches were compared with the method. These methods contained five traditional techniques (non-deep learning) and three deep learning methods. In the following experiments, to ensure the comparison of fairness, the optimal parameters of each approach were used with the default suggested value in the relevant papers of the chosen comparison methods. As shown in Table II, the optimal parameters for each method (including the proposed approach in this article) are comprehensively listed in the table.

Six evaluation metrics also were employed to measure the performance of the method on the LIM task, including three positive indicators and three negative indicators, namely, overall accuracy (OA), average accuracy (AA), kappa coefficient (KA), false alarm (FA) rate, miss alarm (MA) rate, and total error (TE). These evaluation metrics are all calculated based on the confusion matrix, which comprises four parameters: true negative (TN), false negative (FN), true positive (TP), and false positive (FP). The confusion matrix is currently the most common and widely used tool in image accuracy evaluation. Details of these evaluation indicators are shown in Table III.

C. Experimental Comparisons and Visual Inspection

The experimental results were compared and presented in the form of visual performance and quantitative tables. As shown in Figs. 3 and 4, the visual performance and comparisons indicated that the change detection maps achieved by the proposed approaches have less missed detection and false detection pixels, and less salt-and-pepper noise was obtained at the detection map of the proposed approach. As shown in Figs. 5–7, the "CC," "UC," "FD," and "MD" denoted the correct change detection pixels, the unchanged pixels, the false detection pixels, and the missed detection pixels. Therefore, the shape and size of a landslide area can be better preserved in the change detection of our proposed approach than that of other approaches. For example, as shown in Table IV, the proposed approach achieved an improvement of about 93.14 — 98.95 in terms of OA for Dataset #1. The quantitative comparisons in Tables V and VI well supported the visual performance observation. As shown in Table VII, the proposed approach also achieved the best accuracies in terms of all the evaluation indicators for Dataset #4. Therefore, the experimental results based on comparing eight state-of-the-art methods and four pairs of remote-sensed images indicated that the proposed approach has an advantage in detecting a better landslide inventory map with bitemporal remote-sensed images. Although the proposed approach needs some empirical parameters, compared with the popular deep learning techniques which usually require large amounts of training samples, the proposed approach also has advantages in achieving the satisfactory performance of LIMs in an unsupervised way.

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TABLE III
QUANTITATIVE ACCURACY EVALUATION INDICATORS FOR EACH EXPERIMENT

Evaluation Indicators	Formula	Definition
Overall Accuracy (OA)	$OA = \frac{TP+TN}{TP+TN+FP+FN}$	OA is the correctly detected pixels between the total pixels of the ground map.
Average Accuracy (AA)	AA = $(\frac{TP}{TP+FP} + \frac{TN}{TN+FN})/2$	AA is the mean of correctly detected changed and correctly unchanged ratios.
Kappa Coefficient (Ka)	$P_0 = \frac{TP+TN}{TP+TN+FP+FN}$ $P_e = \frac{(TP+FP)\times(TP+FN)+(FN+TN)\times(FP+TN)}{(TP+TN+FP+FN)^2}$ $Ka = \frac{P_0-P_e}{1-P_e}$	Ka reflects the reliability of the detection map by measuring inter-rater reliability for changed and unchanged classes.
False Alarm (FA)	$FA = \frac{FP}{TN+FP}$	FA is the ratio between false changed and unchanged pixels of ground truth.
Missing Alarm (MA)	$MA = \frac{FN}{TP+FN}$	MA is the ratio between false unchanged and changed pixels of ground truth.
Total Error (TE)	$TE = \frac{FP+FN}{TP+TN+FP+FN}$	TE is the ratio between the summary of false changed and false unchanged pixels and the total pixels of the ground map.
Precision (Pr)	$Pr = \frac{TP}{TP+FP}$	The ratio between the number of changed pixels correctly identified and the total number of changed pixels in the result.

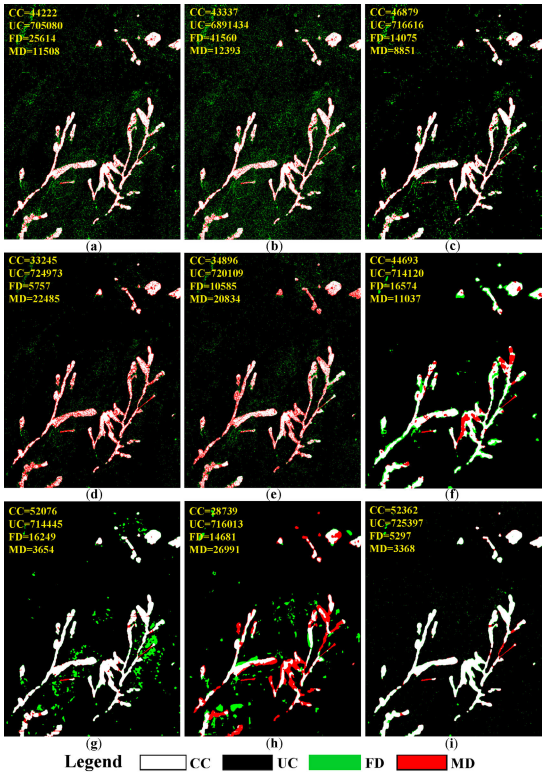


Fig. 4. To obtain landslide inventory maps using different methods for Dataset #1. (a) MLS [28]. (b) CVA_FCM [25]. (c) Semi_FCM [36]. (d) CDMRF [31]. (e) KNN [60]. (f) GWDM [34]. (g) KPCAMNet [64]. (h) DSMANet [58]. (i) Proposed S^{3}AR-based approach.

TABLE IV
QUANTITATIVE EVALUATION AND COMPARISON (%) OF THE SELECTED STATE-OF-THE-ART METHODS WITH THE PROPOSED METHOD ON LIM FOR DATASET #1

Method	OA	AA	Ka	FA	MA	TE
MLS [28]	95.28	87.92	0.68	3.50	20.65	4.72
CVA_FCM [25]	93.14	86.04	0.58	5.69	22.24	6.86
Semi_FCM [36]	97.08	91.10	0.79	1.93	15.88	2.92
CDMRF [31]	96.41	79.43	0.68	0.79	40.35	3.59
KNN [60]	96.00	80.58	0.67	1.45	37.38	4.00
GWDM [34]	96.49	88.96	0.75	2.27	19.80	3.51
KPCAMNet [64]	97.47	95.61	0.82	2.22	6.56	2.53
DSMANet [58]	94.70	74.78	0.55	2.00	48.43	5.30
S^{3}AR _OTSU	**98.95**	**96.59**	**0.92**	**0.66**	**6.17**	**1.05**

TABLE V
QUANTITATIVE EVALUATION AND COMPARISON (%) OF THE SELECTED STATE-OF-THE-ART METHODS WITH THE PROPOSED METHOD ON LIM FOR DATASET #2

Method	OA	AA	Ka	FA	MA	TE
MLS [28]	92.98	88.51	0.51	6.53	16.44	7.02
CVA_FCM [25]	82.94	82.83	0.26	17.05	17.30	17.06
Semi_FCM [36]	90.59	87.98	0.43	9.12	**14.92**	9.41
CDMRF [31]	94.32	59.42	0.24	1.87	79.29	5.68
KNN [60]	96.57	75.54	0.58	**1.13**	47.78	3.43
GWDM [34]	95.16	68.91	0.42	1.98	60.20	4.84
KPCAMNet [64]	92.60	87.67	0.49	6.86	17.80	7.40
DSMANet [58]	96.12	84.47	0.62	2.60	28.46	3.86
S^{3}AR _OTSU	**97.01**	**89.72**	**0.71**	2.20	18.37	**3.00**

D. Discussion

To promote the wide application of the proposed S^{3}AR-based method for achieving the task of LIMs with RSIs, we discussed the relationship between the parameters (T_1, T_2, N) and the detection accuracies of the proposed method.

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TABLE VI

QUANTITATIVE EVALUATION AND COMPARISON (%) OF THE SELECTED STATE-OF-THE-ART METHODS WITH THE PROPOSED METHOD ON LIM FOR DATASET #3

Method	OA	AA	Ka	FA	MA	TE
MLS [28]	83.46	71.05	0.83	13.36	44.54	16.54
CVA_FCM [25]	75.19	71.38	0.24	23.83	33.41	24.81
Semi_FCM [36]	86.20	79.47	0.86	12.07	28.99	13.80
CDMRF [31]	92.60	77.58	0.58	3.54	41.30	7.39
KNN [60]	92.43	77.25	0.57	3.67	41.84	7.57
GWDM [34]	90.05	78.60	**0.90**	7.01	35.79	9.95
KPCAMNet [64]	86.13	85.30	0.48	13.66	15.73	13.87
DSMANet [58]	93.26	86.97	0.67	5.12	20.94	6.74
S^3AR _OTSU	**95.70**	**92.58**	0.78	**3.50**	**11.34**	**3.00**

TABLE VII

QUANTITATIVE EVALUATION AND COMPARISON (%) OF THE SELECTED STATE-OF-THE-ART METHODS WITH THE PROPOSED METHOD ON LIM FOR DATASET #4

Method	OA	AA	Ka	FA	MA	TE
MLS [28]	97.33	95.09	0.81	2.59	**7.23**	2.94
CVA_FCM [25]	97.15	94.03	0.81	2.29	9.65	2.85
Semi_FCM [36]	97.75	95.15	0.85	1.79	7.92	2.25
CDMRF [31]	97.71	87.94	0.82	**0.57**	23.54	2.30
KNN [60]	97.86	91.83	0.84	1.08	15.27	2.15
GWDM [34]	93.24	78.12	0.54	4.09	39.67	6.76
KPCAMNet [64]	92.00	90.91	0.59	7.81	10.38	8.00
DSMANet [58]	97.23	84.12	0.77	0.46	31.31	2.77
S^3AR _OTSU	**98.03**	**95.46**	**0.86**	1.52	7.55	**1.97**

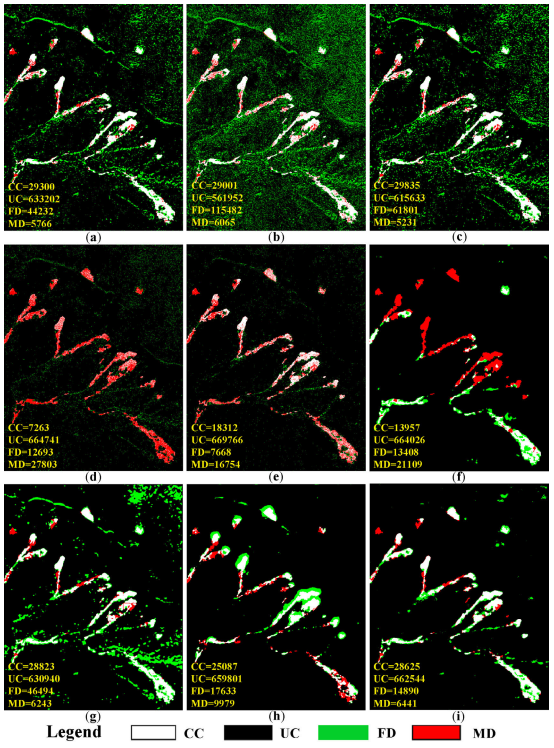


Fig. 5. To obtain landslide inventory maps using different methods for Dataset #2. (a) MLS [28]. (b) CVA_FCM [25]. (c) Semi_FCM [36]. (d) CDMRF [31]. (e) KNN [60]. (f) GWDM [34]. (g) KPCAMNet [64]. (h) DSMANet [58]. (i) Proposed S^3AR-based approach.

First, to investigate the relationship between the parameters (T_1, T_2) and the detection accuracy of the proposed

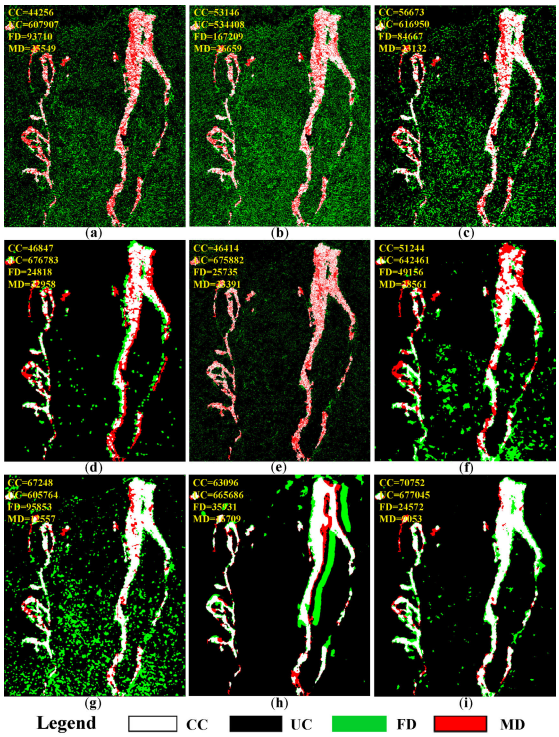


Fig. 6. To obtain landslide inventory maps using different methods for Dataset #3. (a) MLS [28]. (b) CVA_FCM [25]. (c) Semi_FCM [36]. (d) CDMRF [31]. (e) KNN [60]. (f) GWDM [34]. (g) KPCAMNet [64]. (h) DSMANet [58]. (i) Proposed S^3AR-based approach.

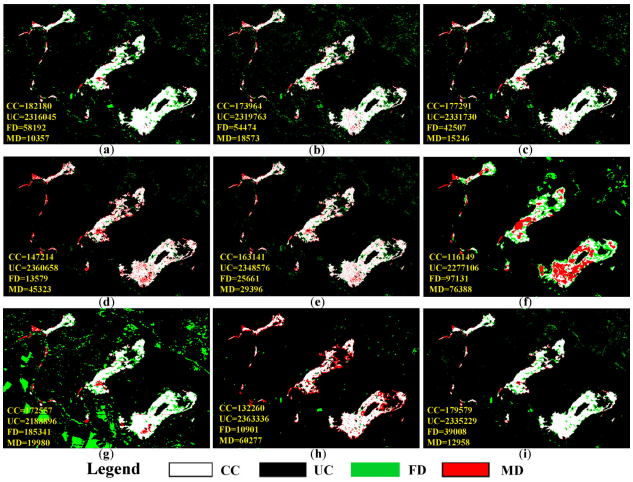


Fig. 7. To obtain landslide inventory maps using different methods for Dataset #4. (a) MLS [28]. (b) CVA_FCM [25]. (c) Semi_FCM [36]. (d) CDMRF [31]. (e) KNN [60]. (f) GWDM [34]. (g) KPCAMNet [64]. (h) DSMANet [58]. (i) Proposed S^3AR-based approach.

S^3AR-based approach, one of the parameters was fixed and the other one was varied to observe the change of detection accuracies. As shown in Fig. 8, it can see that the detection accuracies in terms of OA, AA, and KA were increased gradually with the increase of T_1, then when the accuracies reached the best accuracies in terms of each specific dataset, the accuracies remained at a similar accuracy with the increase of T_1. For example, the accuracy of PE was increased from 19.65 to 67.31 for Dataset-1 when T_1 was increased from 10 to 70 and T_2 was fixed at 150. Then, PE remained between

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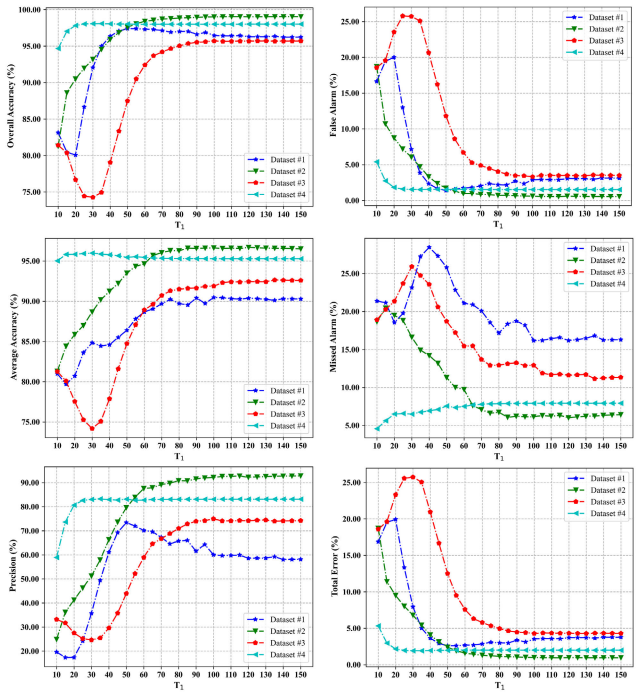


Fig. 8. Relationship between detection accuracies and the change of T_1 when T_2 is fixed (T_2 is fixed at 150, 200, 300, and 150 for Dataset-1, Dataset-2, Dataset-3, and Dataset-4, respectively).

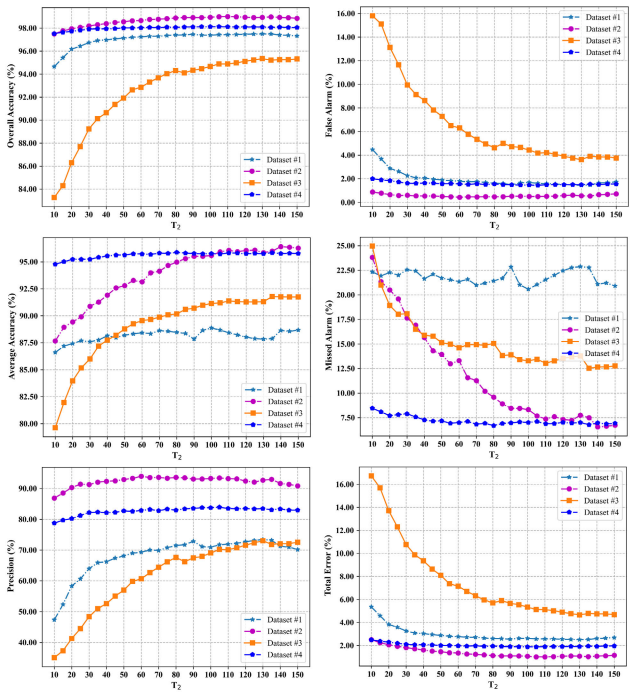


Fig. 9. Relationship between detection accuracies and the change of T_2 when T_1 is fixed (T_1 is fixed at 60, 80, 100, and 40 for Dataset-1, Dataset-2, Dataset-3, and Dataset-4, respectively).

65.78 and 58.12 when T_1 was creased from 80 to 150. A similar phenomenon can be obtained when we observe the other evaluation indicators in terms of other datasets. The reasons can be summarized below: 1) due to T_2 being fixed for each adaptive region, an excessively large T_1 will not make the size of an adaptive region too large and 2) the relationship between detection accuracies and the change of T_1 is different for different datasets because the spatial-uncertainty of each dataset is different.

Second, we observed the relationship between the detection accuracies and the change of T_2 when T_1 is fixed for each dataset, as shown in Fig. 9. As well as T_1, the positive detection accuracies, such as OA, AA, and Pr, are increased with the increase of T_2, and when the detection accuracies reach the top and then they remain at a similar accuracy when T_2 increases continually. For example, OA was increased from 94.65 to 97.4 when T_2 was increased from 10 to 80 for Dataset-1. Then, the detection accuracy of OA remained at between 97.46 and 97.32 when T_2 was increased from 90 to 150. Compared with the relationship between the detection accuracies and the change of T_1, the detection accuracies are not sensitive to the change of T_2 for some datasets.

Third, when we applied our proposed approach to different datasets with different numbers of direction lines, it was found that different datasets reflected different performances to the change of the total number of direction lines. For example, as shown in Fig. 10, the detection accuracies of OA are decreased from 85.17 to 62.86 with the N increasing from 4 to 32 in terms of Dataset-1. When we observed on Dataset-4, it can be seen that the detection accuracies in terms of OA, AA, and Pr are not sensitive to the change of the total number of direction lines, such as OA remained at 97.98 to 97.43 when N was

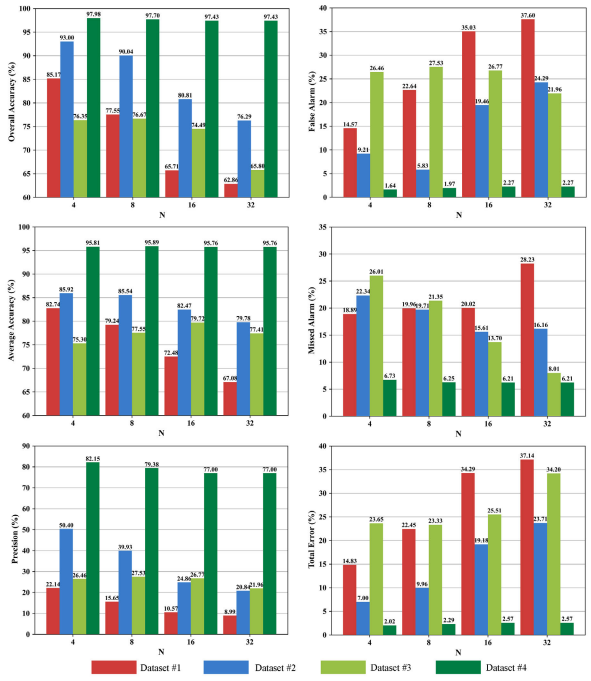


Fig. 10. Relationship between classification accuracies and iterations of proposed approach coupled with SVM for Data-1.

increased from 4 to 32. The relationship between the detection accuracies and the negative evaluation indicators (FA, MA, and TE) well supported the observed phenomenon from the relationship between the detection accuracies and the change of the positive evaluation indicators (OA, AA, and Pr).

From the discussion above, it can be found that 1) an optical setting of T_1 and T_2 may require trial-and-error experiments when applying our proposed approach to a specific dataset

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because different dataset has different spatial uncertainty in terms of shape and sizes of the landslide area, and different spatial contextual information around each pixel is required to smoothen noise and improve the performance of LIMs; and 2) the parameter N denotes the level of shape description, some datasets are sensitive to the change of parameter's setting, and some datasets are not sensitive to the change of parameter's setting, and the sensitive level is different for each evaluation indicator. Therefore, in the application of our proposed approach, the optical setting of T_1, T_2, and N needs a series of experiments for a given dataset in practical applications.

IV. Conclusion

In this article, a novel S^3AR-based approach was proposed to utilize the spatial–spectral contextual information adaptively. In the proposed approach, the shape description vector of an adaptive region is first proposed to depict the shape of an adaptive region; and then a spatial–spectral similarity between a pairwise of adaptive regions was promoted instead of measuring the change magnitude distance between a pairwise pixels directly. Based on the four pairs of RSIs for motorizing the land cover change event caused by landslide and compared with some state-of-the-art methods (MLS [28], CVA_FCM [25], Semi_FCM [36], CDMRF [31], KNN [60], GWDM [34], KPCAMNET [64], and DSMANET [58]), the quantitative evaluation and visual performance well demonstrated the feasibility and the advantage of the proposed S^3 AR-based approach for LIMs with RSIs. Moreover, the proposed approach is an unsupervised approach which may be preferred in practical applications.

Although the proposed method can achieve better visual performance and quantitative evaluation for LIMs with RSIs, the performance of the proposed approach depends on an optical composition of the parameters setting of the proposed approach, the process of which requires trial-and-error experiments which is time-consuming. Therefore, in future studies, some parameter estimation approaches will be studied and combined with our proposed approach to improve the usability of the proposed approach.

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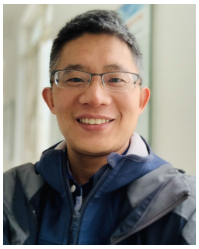
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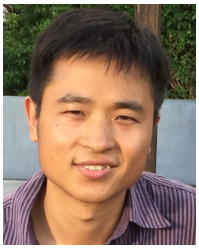
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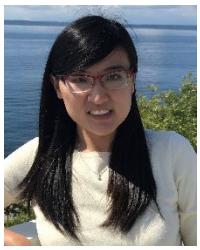
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